

i3 SmartLabs & AI Platform Commercialization Blueprint

Multi-Cloud Infrastructure, 45-Vendor Lab Integration, Credit Balancing, and Revenue Strategy

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Target Environment: IBM Cloud ROKS 4.17 · Red Hat AAP · Hybrid Edge

Date: Q4 2026 Strategic Plan

Ksh 155M YEAR 1 TARGET REVENUE	45 Vendors MULTI-TENANT LAB MATRIX	USD 36K/Yr IBM CLOUD CREDITS (3 ACCOUNTS)	2,500 CONCURRENT STUDENT CAPACITY
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1. EXECUTIVE COMMERCIAL STRATEGY & ECOSYSTEM MONETIZATION

i3 Technologies Limited (Instrumented, Interconnected & Intelligent Technologies Limited), headquartered in Nairobi, operates as an enterprise ICT solutions architect, systems integrator, and premier technical workforce incubator. Our existing production footprint spans 10 operational namespaces on Red Hat OpenShift on IBM Cloud (ROKS 4.17 in Frankfurt), managing enterprise-grade workloads including **EvalOS** (AI certification platform), **AfroERP** (ERPNext v15 with KRA eTIMS & Safaricom Daraja M-Pesa), **Engage** (CRM & multi-channel messaging), **PMaaS** (Political Management as a Service), and **JupyterHub / Sage AI Lab** backed by on-cluster quantized Large Language Models (Granite, Qwen 2.5, Qwen Coder, LLaVA Vision).

By operationalizing **i3 SmartLabs** as a unified Lab-as-a-Service (LaaS) and Exam-as-a-Service (EaaS) commercial engine, i3 Technologies captures multi-tier recurring revenue across four proven streams, projecting an EBITDA of **Ksh 89.95M (58.0% margin)** in Year 1 scaling to **Ksh 287M (71.8% margin)** in Year 3.

COMMERCIAL REVENUE STREAM	TARGET SEGMENT & VALUE PROPOSITION	PRICING MODEL & UNIT ECONOMICS	YEAR 1 TARGET (KSH)
Stream 1: Customer & Corporate Training	Individual learners, universities, and enterprise technical teams requiring industry certifications across 45 vendor tracks.	Individual: Ksh 25,000/seat (3 labs, 100 hrs) Pro: Ksh 75,000/seat (10 labs, exam vouchers) Enterprise: Ksh 150,000/seat (Custom labs + analytics)	Ksh 8.75M (120 enterprise/pro seats)
Stream 2: Labs-as-a-Service (LaaS)	System integrators, software vendors, and enterprise banks needing isolated sandbox environments for testing and customer POCs.	POC Environment: Ksh 100,000/month Dev/Test Infrastructure: Ksh 350,000/month Dedicated Production Mirror: Ksh 500,000/month	Ksh 43.20M - 69.60M (26 active enterprise clients)
Stream 3: Implementation Services	High-margin enterprise deployments unlocked through pre-validated lab proofs-of-concept (POCs) and automated architecture templates.	Network Infra: Ksh 2.5M/project Security Architecture: Ksh 4.0M/project Cloud/DC Virtualization: Ksh 5.0M - 7.5M/project	Ksh 80.00M - 188.00M (40 enterprise engagements)

COMMERCIAL REVENUE STREAM	TARGET SEGMENT & VALUE PROPOSITION	PRICING MODEL & UNIT ECONOMICS	YEAR 1 TARGET (KSH)
Stream 4: Recurring Managed Services	Continuous Day-2 cluster operations, AI model fine-tuning, security patch compliance, and infrastructure telemetry monitoring.	Annual SLA retainers ranging from Ksh 2.5M to Ksh 5.0M per enterprise client.	Ksh 25.00M (10 annual enterprise contracts)

2. UNIFIED LAB GATEWAY ARCHITECTURE & INGRESS FEDERATION

To deliver a frictionless customer experience, i3 SmartLabs eliminates fragmented cloud consoles by implementing a **Single-Pane-of-Glass Hybrid Lab Ingress Gateway**. A unified authentication and policy enforcement plane routes corporate clients, students, and engineers across multiple IBM Cloud VPCs, on-premises Power Systems (750 VMs), and external simulator workloads.

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# Central Ingress Architecture & Session Routing Topology [Internet / Enterprise WAN] | ▼ (HTTPS / WSS
via Cloudflare DDoS Protection & Anycast CDN) i3 Unified Edge Gateway: labs.i3technologies.co.ke (Envoy /
IBM ALB) | ↳ Identity Federation Plane: Red Hat Keycloak SSO (sso.i3technologies.co.ke/auth) | ↳
Realm: i3 | Clients: evalos-spa, jupyterhub, sage, afroerp, engage, pmaas | ↳ OpenBao Vault: Secret
leasing, ephemeral API keys, SSH keys | ↳ Account 1 (ROKS Core Production): evalos_db, i3-data
(Postgres HA PGO), GitOps ArgoCD | ↳ Account 2 (ROKS SmartLabs Engine): JupyterHub Workers,
Containerized Lab Pods (RHEL, Kali, Cisco) | ↳ Account 3 (AI Model & Inference Gateway): LiteLLM (:
4000) + Ollama + Langfuse Tracing ↳ Aliases: granite-nano (2B), qwen-fast (7B), coder (7B), qwen-heavy
(14B), vision (13B)
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Architectural Gateway Principles

- **Single Sign-On & RBAC Enforcement:** All lab users authenticate via the central Keycloak instance (in namespace i3-auth). User groups dynamically trigger role-based access control (RBAC), determining whether a candidate launches an isolated EvalOS coding sandbox, an instructor-led Jupyter notebook, or an administrative terminal.
- **Unified AI Model Gateway via LiteLLM:** Both internal platform co-workers (Zuri, Sage, Mfumo, Nuru, Dawa) and hands-on lab exercises access artificial intelligence exclusively through the centralized LiteLLM proxy at http://litellm-proxy.i3-model-gateway.svc.cluster.local:4000. This unifies CPU-based Ollama models and external watsonx APIs while logging all token consumption to Langfuse for precise per-tenant billing.
- **Ephemeral Session Governance:** Lab containers automatically terminate after 60 minutes of inactivity (via JupyterHub and Kubernetes culling operators), persisting user configurations to personal Ceph/NFS PVCs while releasing unutilized compute back to the cluster pool.

3. 45-VENDOR MULTI-TENANT LAB INTEGRATION MAP

i3 SmartLabs integrates the entire enterprise vendor landscape into containerized, virtualized, and emulated runtimes running on Red Hat OpenShift and IBM Power Systems:

TECHNOLOGY DOMAIN	VENDORS INCLUDED	VIRTUAL RUNTIMES & SIMULATION ENGINES	CERTIFICATION & INDUSTRY BADGES	NAMESPACE & ISOLATION MODE
Core Networking	Cisco, Juniper, Aruba Networks, Arista Networks	Cisco VIRL/CML, IOSv, Juniper vSRX/vMX, Aruba CX Simulator, Arista vEOS	CCNA (200-301), CCNP Enterprise, JNCIA-Junos, ACMA, ACE	labs-networking Calico SDN Egress isolation
Cybersecurity & Firewalls	Fortinet, Palo Alto, Check Point, Sophos, CrowdStrike	FortiGate VM, FortiAnalyzer, Palo Alto VM-Series, Check Point CloudGuard, Kali Linux	NSE 4/5/7, PCNSE, CCSA/CCSE R81, Sophos Certified Engineer, CEH	labs-security Internal tap interface

TECHNOLOGY DOMAIN	VENDORS INCLUDED	VIRTUAL RUNTIMES & SIMULATION ENGINES	CERTIFICATION & INDUSTRY BADGES	NAMESPACE & ISOLATION MODE
Multi-Cloud & Hyperscalers	IBM Cloud, AWS, Microsoft Azure, Google Cloud (GCP)	LocalStack Pro, Azure Resource Manager Simulator, GCP Emulator, IBM Cloud CLI S2I	AWS Solutions Architect, Azure AZ-104/AZ-305, GCP Cloud Engineer, IBM watsonx	labs-cloud Ephemeral credential injection
Enterprise Systems & Virtualization	IBM Power Systems, VMware, Nutanix, Red Hat	IBM AIX & IBM i LPARs (0.5 core, 4GB RAM), Nested ESXi 8.0, KubeVirt OpenShift Virt	IBM Power Systems Specialist, VMware VCP-DCV, RHCSA, RHCE, OpenShift EX280	labs-virtualization Direct Link to Power pool
Data, Backup & Observability	Oracle, Splunk, Dynatrace, Veeam, F5 BIG-IP	Oracle Database Free / 19c containers, Splunk Enterprise testbed, BIG-IP Virtual Edition	Oracle OCP, Splunk Certified Power User, F5 CTS, Veeam VMCE	labs-data-observability Dedicated NFS volume

4. IBM CLOUD CREDIT OPTIMIZATION & MULTI-ACCOUNT BALANCING

i3 Technologies possesses **three distinct IBM Cloud accounts**, each credited with **USD 3,000 per quarter starting Q4** (aggregating to **USD 9,000/quarter** or **USD 36,000 annually**). To maximize efficiency and prevent credit lapse or cash outlays, workloads are strictly apportioned across the accounts:

IBM CLOUD ACCOUNT	TARGET WORKLOAD & ARCHITECTURE ROLE	MONTHLY ALLOCATION	QUARTERLY BUDGET	OPTIMIZATION & SCALING CONTROLS
Account 1: Production Core & Platform SaaS	Hosts revenue-generating production services: EvalOS Web, AfroERP, Engage, Keycloak SSO, Postgres HA (Crunchy PGO), and ArgoCD GitOps engine.	\$1,000 / mo	\$3,000 / Qtr	Fixed 3-node ROKS cluster (bx2-4x16); BUYVPC partner tier (60% discount). Baseline operational SLA of 99.9%.
Account 2: i3 SmartLabs Training & Compute Engine	Runs high-concurrency student lab workloads: JupyterHub notebooks, containerized Linux/Kali sandboxes, LocalStack, and network node emulators.	\$1,000 / mo	\$3,000 / Qtr	KEDA cron autoscaling: scales up to 8 worker nodes during business/bootcamp hours (08:00–20:00 EAT) and scales to baseline overnight.
Account 3: Model Gateway & GPU Sandbox Labs	Houses LiteLLM model proxy, Langfuse tracing, specialized watsonx AI developer sandboxes, and burstable GPU Virtual Server Instances (NVIDIA A100).	\$1,000 / mo	\$3,000 / Qtr	Leverages GPU4YOU 50% discount voucher (\$1.80/hr per A100 PCIe) to provide high-speed model fine-tuning without exhausting budget.
Consolidated Balance	Tri-Account Federated SmartLabs Cloud Fabric	\$3,000 / mo	\$9,000 / Qtr	Zero cash infrastructure OpEx for initial ramp-up.

5. LICENSING STRATEGY: ZERO-COST RUNTIMES & PARTNER ENTITLEMENTS

Achieving aggressive gross margins requires eliminating commercial licensing friction during platform ramp-up. We leverage established vendor partner channels, developer licenses, and open-source foundations:

- **Red Hat Developer for Organizations & Partner Portal:** Grants no-cost enterprise subscriptions for up to 16 RHEL instances, complimentary OpenShift Container Platform development tiers, and Not-For-Resale (NFR) licenses for Ansible Automation Platform (AAP) and Red Hat Advanced Cluster Management (RHACM).

- **IBM Partner Plus Software Access Catalog:** Access to full software production-mirror licenses including IBM Db2 SaaS, IBM MQ, API Connect, and watsonx Orchestrate/Assistant practice licenses for demonstration and bootcamp enablement.
- **Vendor NFR Program Activation:** Formal enrollment into Fortinet Engage (FortiGate VM/FortiAnalyzer), Palo Alto NextWave (VM-Series firewalls), Check Point Partner Program (CloudGuard), and VMware Partner Connect (vSphere/ ESXi NFRs) to equip 45 vendor lab pods with legitimate testing licenses.
- **Cloud Simulator Frameworks:** Utilizing LocalStack Pro (AWS), Azure Resource Manager local emulator, and GCP Google Cloud SDK test suites to deliver authentic cloud API environments with zero external egress or cloud consumption billing.
- **Open-Source Core Stack:** Kubernetes-native open-source building blocks: Crunchy Data PGO (PostgreSQL HA), Ollama + LiteLLM (LLM serving), Monaco Editor (VS Code in-browser engine), and OpenBao (Vault-compatible secret management).

Capital Expenditure Analysis for Ramp-Up: Software licensing cost during Phase 1 is **Ksh 0 (\$0)** through full utilization of Partner NFRs and Developer subscriptions. Compute costs are **100% offset** by the \$36,000/yr IBM Cloud Credits. Total cash investment required is limited to marketing, exam voucher procurement, and staff payroll.

6. INFRASTRUCTURE AUTOMATION: RED HAT ANSIBLE PLAYBOOK FRAMEWORK

Manual lab configuration does not scale to 2,500 simultaneous users. Red Hat Ansible Automation Platform (AAP) orchestrates end-to-end lifecycle provisioning, configuration drift repair, and scheduled teardown across our multi-cloud fabric.

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--- # i3 SmartLabs Automated Student Sandbox Provisioner - name: Provision Multi-Vendor Candidate Lab Environment hosts: localhost gather_facts: no vars: student_id: "std-kenya-0842" lab_track: "cisco-fortinet-hybrid" session_ttl: "14400" # 4 hours tasks: - name: Create Isolated OpenShift Namespace kubernetes.core.k8s: name: "lab-{{ student_id }}" kind: Namespace state: present - name: Apply Strict Network Policy & Resource Quota kubernetes.core.k8s: definition: apiVersion: v1 kind: ResourceQuota metadata: name: student-quota namespace: "lab-{{ student_id }}" spec: hard: requests.cpu: "2" requests.memory: "4Gi" limits.cpu: "4" limits.memory: "8Gi" - name: Deploy FortiGate Virtual Appliance & Linux Terminal kubernetes.core.k8s: state: present src: "roles/smartlabs/templates/fortigate-pod.yaml.j2" - name: Schedule Automated Teardown via KEDA/AAP Callback ansible.builtin.shell: cmd: "echo 'oc delete ns lab-{{ student_id }}' | at now + 4 hours"
```

7. COMMERCIALIZATION OF PLATFORM APPLICATIONS

PLATFORM APPLICATION	TECHNICAL ARCHITECTURAL PROFILE	COMMERCIAL PACKAGING & TARGET MARKET	PROJECTED REVENUE TRAJECTORY
EvalOS (Exam & Skills Testing)	Next.js 14, Monaco code sandbox, Crunchy Postgres HA, LiteLLM with Qwen 7B Study Coach, and Zuri AI assistant.	Exam-as-a-Service (EaaS): B2B talent screening for tech banks and recruitment agencies. B2C practice exams for AWS, Red Hat, and IBM certifications. <i>Pricing: Ksh 1,500 - 3,500/test or Ksh 65,000/mo enterprise subscription.</i>	Year 1: Ksh 18.5M Year 2: Ksh 42.0M Year 3: Ksh 75.0M
AfroERP (Enterprise ERP & Tax)	ERPNext v15, MariaDB, Redis, integrated Safaricom Daraja STK Push / B2C, real-time KRA eTIMS invoice generation, Mfumo AI.	SME Compliance SaaS: All-in-one ERP for Kenyan retail, supply chain, and manufacturing businesses needing automated eTIMS & M-Pesa ledger sync. <i>Pricing: Ksh 6,500 - 18,500/month per business tenant.</i>	Year 1: Ksh 14.2M Year 2: Ksh 35.0M Year 3: Ksh 60.0M

PLATFORM APPLICATION	TECHNICAL ARCHITECTURAL PROFILE	COMMERCIAL PACKAGING & TARGET MARKET	PROJECTED REVENUE TRAJECTORY
Engage (Customer Communications)	Node.js/Express, Africa's Talking SMS API, Meta WhatsApp Business API, Brevo Email, Nuru AI copywriter.	Conversational CRM: Automated transactional alerts, two-way WhatsApp marketing, and payment collection reminders. <i>Pricing: Ksh 5,000/mo base platform + wholesale pass-through messaging tariffs.</i>	Year 1: Ksh 12.0M Year 2: Ksh 28.0M Year 3: Ksh 45.0M
PMaaS (Campaign Management)	NextAuth, Keycloak, Qwen 14B strategic briefing engine, voter constituency mapping, mass M-Pesa field disbursements, Dawa AI co-worker.	Political Campaign-as-a-Service: Turnkey digital operations for parliamentary, gubernatorial, and civic election campaigns across East Africa. <i>Pricing: Ksh 200,000 - 750,000/month per campaign retainer.</i>	Year 1: Ksh 25.0M Year 2: Ksh 55.0M Year 3: Ksh 80.0M
AI Lab / JupyterHub (Data Science Workbench)	JupyterHub, Open WebUI (Sage), ChromaDB vector store, on-cluster LiteLLM gateway with granite-nano, qwen-fast, and coder.	AI Workforce Incubator: Turnkey data science sandboxes for corporate upskilling, bootcamps, and hackathons. <i>Pricing: Ksh 4,000/student/mo or Ksh 350,000 per corporate training cohort.</i>	Year 1: Ksh 15.0M Year 2: Ksh 30.0M Year 3: Ksh 50.0M

8. 12-MONTH FINANCIAL PROJECTIONS & 90-DAY EXECUTION ROADMAP

Consolidating our multi-tenant vendor matrix, IBM Cloud credit allocation, and application revenue streams yields the following 3-year P&L outlook (in Kenyan Shillings):

FINANCIAL METRIC (KSH)	YEAR 1 (FOUNDATION)	YEAR 2 (EXPANSION)	YEAR 3 (MARKET SCALE)
Customer & Corporate Training	6,750,000	15,000,000	25,000,000
Labs-as-a-Service (LaaS)	43,200,000	80,000,000	120,000,000
Implementation & Deployment Projects	80,000,000	120,000,000	180,000,000
Recurring Managed Services & SLAs	25,000,000	50,000,000	75,000,000
TOTAL GROSS REVENUE	154,950,000	265,000,000	400,000,000
Infrastructure & Cloud Operations	15,000,000	20,000,000	25,000,000
Software Licenses & Exam Vouchers	12,000,000	15,000,000	18,000,000
Engineering & Instruction Personnel (10-20 staff)	25,000,000	35,000,000	45,000,000
Sales, Marketing & Partner Enablement	8,000,000	12,000,000	15,000,000
Operations, Admin & Facilities Support	5,000,000	8,000,000	10,000,000
TOTAL OPERATING EXPENSES	65,000,000	90,000,000	113,000,000
EBITDA (Operating Profit)	89,950,000	175,000,000	287,000,000
EBITDA Margin %	58.05%	66.04%	71.75%

90-Day Tactical Execution Plan

- **Days 1–30 (Infrastructure Unification & Federated Gateway Setup):** Deploy the central Envoy/ALB gateway fronting Keycloak (``i3-auth``) and LiteLLM (``i3-model-gateway``). Configure cross-account VPC peering between the 3 IBM Cloud

accounts. Partition workloads to begin burning the \$9,000 quarterly credit. Re-link OpenBao for dynamic credential rotation.

- **Days 31–60 (Partner NFR Licensing & Automation Engine):** Enroll into Cisco, Fortinet, Palo Alto, and VMware partner portals to pull NFR virtual images. Deploy the Red Hat Ansible Automation Platform playbook repository to automate student namespace provisioning and cleanup. Complete live Safaricom M-Pesa Daraja and KRA eTIMS credentials in AfroERP and PMaaS secrets.
- **Days 61–90 (Go-to-Market Commercial Launch):** Onboard 10 enterprise pilot partners across financial services, telecom, and higher education. Launch public certification cohorts for AWS, Red Hat, and IBM watsonx on EvalOS. Roll out commercial marketing campaigns via Engage SMS/WhatsApp for AfroERP and SmartLabs subscriptions.

Document Approved for Board Review & Implementation • i3 Technologies Limited • Executive Management Committee